Computing Distance and Midpoint from Coordinates Answer Key

High School Geometry

Quick Answers

1. 5
2. 5, 2
3. 13
4. -1, 1
5. $6\sqrt{2}$
6. 3, 5
7. 4, 1, 5
8. 4, 3.5, 4, 3.5

Common wrong answers

1. *If they got 7: added the horizontal and vertical legs instead of using the Pythagorean form*
3. *If they got 17: added the horizontal and vertical legs instead of using the Pythagorean form*
5. *If they got 12: added the two legs instead of using the Pythagorean form*

1. Distance from $A(1,2)$ to $B(4,6)$.

Solution: The horizontal leg is $4-1=3$ and the vertical leg is $6-2=4$, so the segment is the hypotenuse of a 3–4 right triangle.

$$\begin{aligned} AB &= \sqrt{(4-1)^2+(6-2)^2} \\ &= \sqrt{3^2+4^2} = \sqrt{9+16} = \sqrt{25} \end{aligned}$$

$$\boxed{AB = 5}$$

2. Midpoint of $A(2,-3)$ and $B(8,7)$.

Solution: Average each coordinate separately — the x's with the x's and the y's with the y's.

$$M = \left(\frac{2+8}{2},\ \frac{-3+7}{2}\right) = \left(\frac{10}{2},\ \frac{4}{2}\right)$$

$$\boxed{M = (5,\ 2)}$$

Sanity check: 5 is halfway between 2 and 8, and 2 is halfway between -3 and 7.

3. Distance from $A(-3, 1)$ to $B(2, 13)$.

Solution: Subtracting a negative adds: the horizontal leg is $2 - (-3) = 5$, and the vertical leg is $13 - 1 = 12$.

$$\begin{aligned} AB &= \sqrt{(2-(-3))^2 + (13-1)^2} \\ &= \sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} \end{aligned}$$

$$\boxed{AB = 13}$$

This is the 5–12–13 triple.
Watch for: an answer of 17 — that is 5+12, adding the legs instead of using the Pythagorean relationship between them.

4. Midpoint of $A(-5, 4)$ and $B(3, -2)$.

Solution:

$$M = \left(\frac{-5+3}{2},\ \frac{4+(-2)}{2}\right) = \left(\frac{-2}{2},\ \frac{2}{2}\right)$$

$$\boxed{M = (-1,\ 1)}$$

A negative x with a positive y places the midpoint in Quadrant II.

5. Exact distance from $A(0, 0)$ to $B(6, 6)$.

Solution: Both legs measure 6, so this is an isosceles right triangle and the exact answer is a radical, not a decimal.

$$\begin{aligned} AB &= \sqrt{(6-0)^2 + (6-0)^2} = \sqrt{36 + 36} = \sqrt{72} \\ &= \sqrt{36 \cdot 2} = \sqrt{36}\,\sqrt{2} \end{aligned}$$

$$\boxed{AB = 6\sqrt{2}}$$

(That is about 8.49, but the problem asked for exact radical form.)

6. Find B given $M(3, 5)$ and $A(-1, 2)$.

Solution: The midpoint formula is two averaging equations. Write each one and solve for the missing coordinate.

$$\begin{aligned} \frac{-1 + x_B}{2} &= 3 \implies -1 + x_B = 6 \implies x_B = 7 \\ \frac{2 + y_B}{2} &= 5 \implies 2 + y_B = 10 \implies y_B = 8 \end{aligned}$$

$$\boxed{B = (7,\ 8)}$$

Check by recomputing the midpoint of $A(-1, 2)$ and $B(7, 8)$: $\left(\frac{-1+7}{2},\ \frac{2+8}{2}\right)$, which gives the given midpoint $\boxed{M = (3,\ 5)}$ ✓

7. Median of triangle PQR: $P(1,1)$, $Q(7,1)$, $R(4,6)$.
(a) Midpoint of $\overline{PQ}$. Both endpoints have $y = 1$, so the segment is horizontal and only the x-coordinate needs averaging.

$$M = \left(\frac{1+7}{2},\ \frac{1+1}{2}\right) = \left(\frac{8}{2}, \frac{2}{2}\right)$$

$$\boxed{M = (4,\ 1)}$$

(b) Length of the median $\overline{RM}$. Now use the distance formula on $R(4,6)$ and $M(4,1)$. Both have $x = 4$, so the horizontal leg is 0 and the segment is vertical.

$$RM = \sqrt{(4-4)^2 + (1-6)^2} = \sqrt{0+25} = \sqrt{25}$$

$$\boxed{RM = 5}$$

The median from R is also the triangle's altitude here, because $\overline{PQ}$ is horizontal and $\overline{RM}$ is vertical.

8. Coordinate proof: diagonals of $ABCD$ with $A(0,0)$, $B(6,2)$, $C(8,7)$, $D(2,5)$.
Solution — computation. Diagonal $\overline{AC}$ joins $A(0,0)$ to $C(8,7)$:

$$\left(\frac{0+8}{2},\ \frac{0+7}{2}\right) = \boxed{(4,\ 3.5)}$$

Diagonal $\overline{BD}$ joins $B(6,2)$ to $D(2,5)$:

$$\left(\frac{6+2}{2},\ \frac{2+5}{2}\right) = \boxed{(4,\ 3.5)}$$

Model proof (open response — grade the reasoning). The two diagonals have the same midpoint, $(4, 3.5)$. A midpoint of a segment is by definition the point that divides it into two congruent halves, so that shared point cuts $\overline{AC}$ into two equal parts and cuts $\overline{BD}$ into two equal parts. Because the point lies on both diagonals at once, it is exactly where they cross. Therefore each diagonal passes through the other's midpoint — that is, the diagonals bisect each other.
Look for: both midpoints computed and stated to be equal; the word *bisect* tied to the definition of midpoint; the observation that a common midpoint is the intersection point. A student who computes only one midpoint has not proved anything.